

Installation Guide (Ubuntu)

This guide covers installing the Security Monitoring Dashboard on Ubuntu 20.04 / 22.04 / 24.04. It works on a VPS, a home server, or a laptop.

1. Prerequisites

- Ubuntu 20.04+ (Debian-based distros should also work)
- A user with `sudo` privileges
- Outbound internet access to install packages

The dashboard integrates these tools (installed automatically with `--tools`):

- **nmap** — scanning engine for the offensive toolkit
 - **fail2ban** — intrusion detection / automated banning
 - **ufw** — uncomplicated firewall (or use `iptables`)
-

2. Get the code

Copy the `security_dashboard/` directory to your server, e.g. via `scp`, `git`, or the download button in this UI:

```
# example with scp from your laptop
scp -r security_dashboard user@your-server:/opt/

# on the server
cd /opt/security_dashboard
```

3. Automated install (recommended)

The setup script handles system packages, a Python virtual environment, and (optionally) the security tools and a systemd service.

```
chmod +x scripts/setup.sh

# Option A – full install: python deps + nmap/fail2ban/ufw + systemd service
sudo ./scripts/setup.sh --all

# Option B – tools but no service (you run it manually)
sudo ./scripts/setup.sh --tools

# Option C – minimal: just the Python environment
./scripts/setup.sh
```

What each flag does:

| Flag | Action |

```
|—|—|
| (none) | Installs python3/venv/pip + creates .venv + installs Python requirements |
| --tools | Also installs and enables nmap, fail2ban, ufw |
| --service | Also installs + enables a systemd service ( security-dashboard ) |
| --all | --tools + --service |
```

4. Manual install (alternative)

```
sudo apt-get update
sudo apt-get install -y python3 python3-venv python3-pip nmap fail2ban ufw

python3 -m venv .venv
source .venv/bin/activate
pip install -r requirements.txt

python run.py           # starts on http://0.0.0.0:5000
```

5. First login

1. Open `http://<server-ip>:5000` (or `http://localhost:5000` locally).
2. Log in with the default credentials: `admin / changeme`.
3. **Immediately change the password** — the dashboard offers a change-password action; or edit `config/config.yaml` (`auth.default_password`) **before** first launch and delete `data/users.json` to re-seed.

The password is stored as a salted PBKDF2-SHA256 hash in `data/users.json` (file mode `600`).

6. Configure

Edit `config/config.yaml` (`./config/config.yaml`). At minimum, review:

```
server:
  secret_key: "<set a long random value>" # or SECDASH_SERVER_SECRET_KEY

firewall:
  backend: "ufw" # or "iptables"
  enforce: false # set true to apply REAL firewall rules

offensive:
  enabled: true
  target_allowlist: ["192.168.1.0/24", "203.0.113.10"] # YOUR assets/targets
```

Generate a strong secret key:

```
python3 -c "import secrets; print(secrets.token_hex(32))"
```

Restart after config changes (`systemctl restart security-dashboard` or `restart python run.py`).

7. Enable fail2ban (recommended for defense)

```
sudo cp config/fail2ban-sshd.local.example /etc/fail2ban/jail.local
sudo systemctl restart fail2ban
sudo fail2ban-client status          # verify jails
```

The dashboard auto-detects active jails under **IDS / fail2ban**.

8. Enable the firewall

```
sudo ufw allow OpenSSH          # don't lock yourself out!
sudo ufw allow 5000/tcp         # dashboard (restrict to your IP if possible)
sudo ufw enable
```

Then set `firewall.enforce: true` in `config.yaml` so the dashboard's block buttons apply real `ufw` rules.

9. Run as a service (production)

If you used `--service`, the dashboard runs under `systemd` via `gunicorn`:

```
sudo systemctl status security-dashboard
sudo systemctl restart security-dashboard
sudo journalctl -u security-dashboard -f    # live logs
```

The service runs with a single worker + threads so the in-memory monitoring state stays consistent. To run as root (full features), the template already sets the invoking user — edit `User=` in `/etc/systemd/system/security-dashboard.service` to `root` if desired, then `sudo systemctl daemon-reload && sudo systemctl restart security-dashboard`.

10. Secure remote access

The dashboard speaks plain HTTP. For remote use, **put it behind HTTPS**:

- **Recommended:** reverse-proxy with `nginx` + Let's Encrypt (TLS termination), proxy to `127.0.0.1:5000`, and restrict by IP / VPN.
- **Quick & private:** access over an SSH tunnel:

```
bash
```

```
ssh -L 5000:localhost:5000 user@your-server
```

```
# then browse http://localhost:5000 on your laptop
```

Never expose port 5000 directly to the internet without TLS + authentication + IP allow-listing.

Troubleshooting

Symptom	Fix
"Access denied reading connections" in logs	Run as root / via systemd for full visibility.
Firewall blocks don't apply	Set <code>firewall.enforce: true</code> and run as root.
<code>nmap</code> is not installed	<code>sudo apt-get install nmap</code> or run setup with <code>--tools</code> .
IDS shows "fail2ban not reachable"	Install & start fail2ban; the built-in SSH watcher still runs.
Can't log in	Delete <code>data/users.json</code> to re-seed from config, then restart.
Port 5000 in use	Change <code>server.port</code> in config or <code>SEC-DASH_SERVER_PORT=8080</code> .